

Project Initialization and Planning Phase — Project Proposal Report

Date	04 July 2026
Team ID	SOLO-CCFD-2026
Project Name	Credit Card Fraud Detection using Machine Learning
Maximum Marks	5 Marks

Project Overview

	Automatically detect fraudulent credit card transactions in real time using machine learning, reducing reliance on slow manual review and limiting financial losses.
Scope	Covers the full pipeline: data preprocessing of anonymized transaction features, training/comparing 4 classifiers, and deploying the best model via a Flask web app and IBM Watson ML for real-time and batch scoring.

Problem Statement

	Manual fraud review is slow, error-prone, and cannot keep pace with the volume and subtlety of modern transaction fraud, which occurs in under 0.2% of transactions.
Impact	Undetected fraud causes direct financial loss and erodes cardholder trust; over-flagging genuine transactions frustrates customers and wastes analyst time.

Proposed Solution

	Train and compare Logistic Regression, Decision Tree, Random Forest, and XGBoost on SMOTE-balanced transaction data; select the best model by recall/AUPRC (given the extreme class imbalance) and deploy it for real-time scoring.
Key Features	- Automated real-time fraud/genuine prediction per transaction - Batch CSV screening for compliance review - Cloud deployment via IBM Watson ML for scalability - Simple analyst-facing Flask web UI

Resource Requirements

Resource Type	Description	Specification/Allocation
Computing Resources	CPU/GPU for training the 4 models	Standard CPU (no GPU required for this dataset size)
Memory	RAM for handling ~285k row dataset in memory	8 GB
Storage	Disk space for dataset, trained model file, and logs	2 GB
Frameworks	Python web framework	Flask
Libraries	ML & data libraries	scikit-learn, XGBoost, pandas, numpy, imbalanced-learn, matplotlib, seaborn
Development Environment	IDE / notebook	Jupyter Notebook, VS Code
Data Source, size, format	Kaggle ULB Credit Card Fraud dataset	284,807 rows, csv